

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks:50

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I SANDHIYA S (192521224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SANDHIYA S

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64** Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.

- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64** • Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

DEBUGGING

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

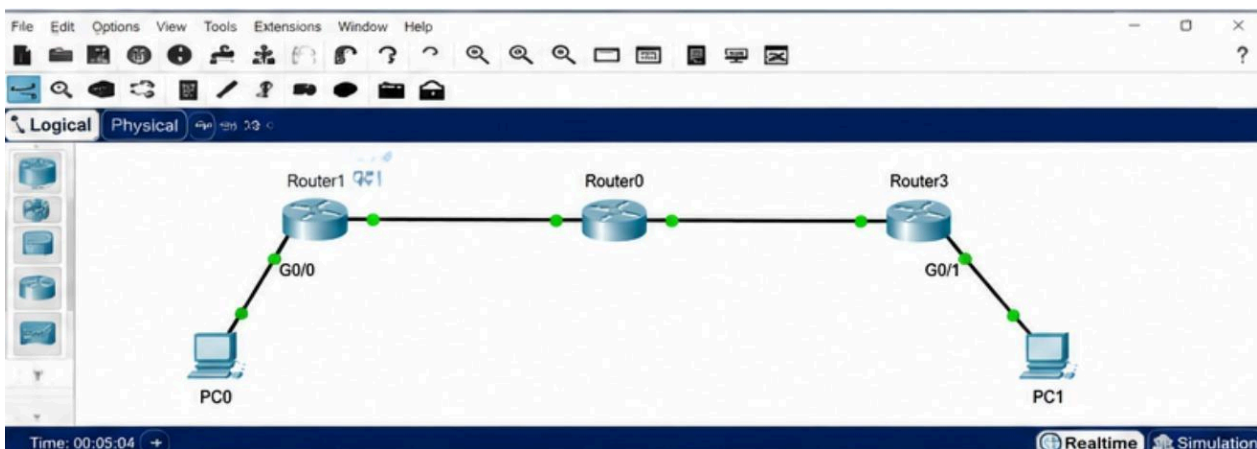
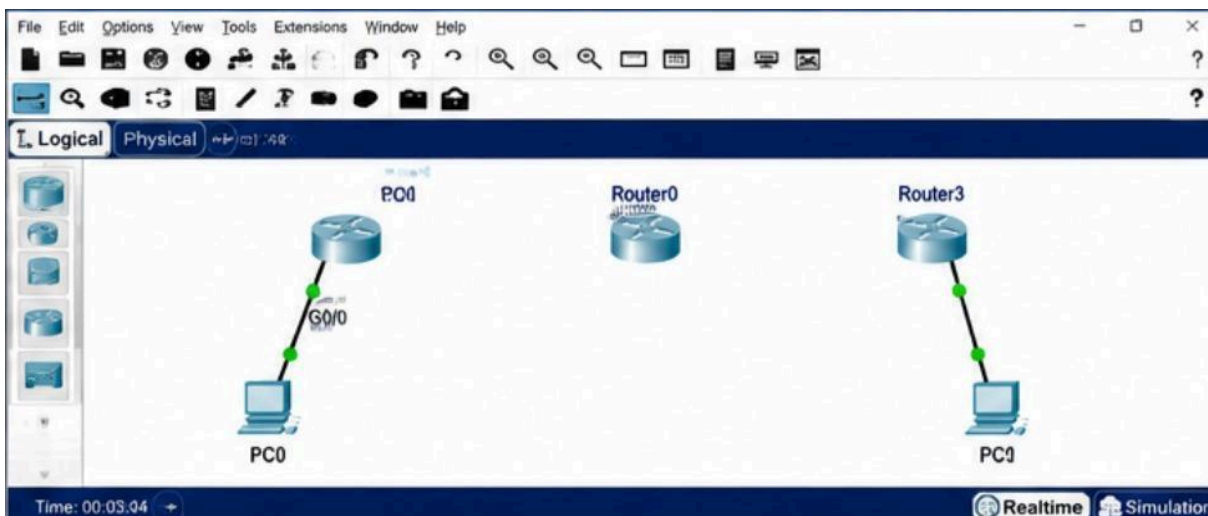
Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

OUTPUT



Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical

Physical

794, y: 287

PC0

192.168.1.10

Fa0

192.168.1.1

G0/0

Router1

G0/1

10.0.0.2

G0/0

Router0

G0/1

10.0.0.4

G0/0

Router3

G0/1

192.168.2.1

Fa0

PC1

192.168.2.100

PC0

Physical Config Desktop Programming Attributes

Command Prompt

Packet Tracer PC Command Line 1.0

PC> ping 192.168.2.100

Pinging 192.168.2.100 with 32 bytes of data:

Reply from 192.168.2.100: bytes=32 time<1ms TTL=126

Reply from 192.168.2.100: bytes=32 time<1ms TTL=126

Reply from 192.168.2.100: bytes=32 time<1ms TTL=126

Reply from 192.168.2.100: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>

Time: 00:03:45.013

Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Event List

Vis.	Time(sec)	Last Device	At Device	Type	Info
👁	0.000	PC0	Router1	ICMP	
👁	0.001	Router1	Router0	ICMP	
👁	0.002	Router0	Router3	ICMP	
👁	0.003	Router3	PC1	ICMP	

Router1

192.168.1.1/24

G0/0

PC0

192.168.1.10

Mask: 255.255.255.0

GW: 192.168.1.1

Router1

G0/1

10.0.0.2/24

Router0

G0/0

10.0.0.1

G0/1

10.0.0.4/24

Router3

G0/0

10.0.0.3

G0/1

192.168.2.1/24

PC1

192.168.2.100

Mask: 255.255.255.0

GW: 192.168.2.1

IP ADDRESS SUMMARY

Device	Interface	IP Address	Subnet Mask
Router1	G0/0	192.168.1.1	255.255.255.0
Router1	G0/1	10.0.0.2	255.255.255.0
Router0	G0/0	10.0.0.1	255.255.255.0
Router0	G0/1	10.0.0.4	255.255.255.0
Router3	G0/0	10.0.0.3	255.255.255.0
Router3	G0/1	192.168.2.1	255.255.255.0
PC0	Fa0	192.168.1.10	255.255.255.0
PC1	Fa0	192.168.2.100	255.255.255.0

Time: 00:07:15

Realtime Simulation

CO3 - AT 3
Computer Networks.

Routing Table Debugging

Objective

The Objective of the experiment is to analyze the routers routing table, determine the appropriate route for the destination IP address **192.168.2.100**. Verify whether the destination belongs to the correct subnet, identify Potential routing table for efficiently Packet for forwarding using Cisco Packet tracer.

Routing Table.

Destination Network	Next Hop.
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route	10.0.0.1.

Analysis

The IP **192.168.2.100** belong to **192.168.2.0/24** and using longest prefix match, this router selects the route and forwards the packet to 10.0.0.3. If delivery fails, Possible issues include incorrect next hop configuration, interface failure, missing return route or subnet errors.

Conclusion.

192.168.2.100 is routed via 10.0.0.3 through 192.168.2.0/24. If no match exist, the default route applies. Proper configuration ensures efficient routing.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, wellstructured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation